

```
In [1]: pip install jupyter-dash dash plotly pandas scikit-learn
```

Requirement already satisfied: jupyter-dash in c:\users\hp\anaconda3\lib\site-packages (0.4.2)

Requirement already satisfied: dash in c:\users\hp\anaconda3\lib\site-packages (2.17.1)

Requirement already satisfied: plotly in c:\users\hp\anaconda3\lib\site-packages (5.9.0)

Requirement already satisfied: pandas in c:\users\hp\anaconda3\lib\site-packages (2.0.3)

Requirement already satisfied: scikit-learn in c:\users\hp\anaconda3\lib\site-packages (1.3.0)

Requirement already satisfied: requests in c:\users\hp\anaconda3\lib\site-packages (from jupyter-dash) (2.31.0)

Requirement already satisfied: flask in c:\users\hp\anaconda3\lib\site-packages (from jupyter-dash) (2.2.2)

Requirement already satisfied: retrying in c:\users\hp\anaconda3\lib\site-packages (from jupyter-dash) (1.3.4)

Requirement already satisfied: ipython in c:\users\hp\anaconda3\lib\site-packages (from jupyter-dash) (8.15.0)

Requirement already satisfied: ipykernel in c:\users\hp\anaconda3\lib\site-packages (from jupyter-dash) (6.25.0)

Requirement already satisfied: ansi2html in c:\users\hp\anaconda3\lib\site-packages (from jupyter-dash) (1.9.2)

Requirement already satisfied: nest-asyncio in c:\users\hp\anaconda3\lib\site-packages (from jupyter-dash) (1.5.6)

Requirement already satisfied: Werkzeug<3.1 in c:\users\hp\anaconda3\lib\site-packages (from dash) (2.2.3)

Requirement already satisfied: dash-html-components==2.0.0 in c:\users\hp\anaconda3\lib\site-packages (from dash) (2.0.0)

Requirement already satisfied: dash-core-components==2.0.0 in c:\users\hp\anaconda3\lib\site-packages (from dash) (2.0.0)

Requirement already satisfied: dash-table==5.0.0 in c:\users\hp\anaconda3\lib\site-packages (from dash) (5.0.0)

Requirement already satisfied: importlib-metadata in c:\users\hp\anaconda3\lib\site-packages (from dash) (6.0.0)

Requirement already satisfied: typing-extensions>=4.1.1 in c:\users\hp\anaconda3\lib\site-packages (from dash) (4.13.2)

Requirement already satisfied: setuptools in c:\users\hp\anaconda3\lib\site-packages (from dash) (68.0.0)

Requirement already satisfied: tenacity>=6.2.0 in c:\users\hp\anaconda3\lib\site-packages (from plotly) (8.2.2)

Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\hp\anaconda3\lib\site-packages (from pandas) (2.8.2)

Requirement already satisfied: pytz>=2020.1 in c:\users\hp\anaconda3\lib\site-packages (from pandas) (2023.3.post1)

Requirement already satisfied: tzdata>=2022.1 in c:\users\hp\anaconda3\lib\site-packages (from pandas) (2023.3)

Requirement already satisfied: numpy>=1.21.0 in c:\users\hp\anaconda3\lib\site-packages (from pandas) (1.24.3)

Requirement already satisfied: scipy>=1.5.0 in c:\users\hp\anaconda3\lib\site-packages (from scikit-learn) (1.11.1)

Requirement already satisfied: joblib>=1.1.1 in c:\users\hp\anaconda3\lib\site-packages (from scikit-learn) (1.2.0)

Requirement already satisfied: threadpoolctl>=2.0.0 in c:\users\hp\anaconda3\lib\site-packages (from scikit-learn) (2.2.0)

Requirement already satisfied: Jinja2>=3.0 in c:\users\hp\anaconda3\lib\site-packages (from flask->jupyter-dash) (3.1.2)

Requirement already satisfied: itsdangerous>=2.0 in c:\users\hp\anaconda3\lib\site-packages (from flask->jupyter-dash) (2.0.1)

Requirement already satisfied: click>=8.0 in c:\users\hp\anaconda3\lib\site-packages (from flask->jupyter-dash) (8.0.4)

Requirement already satisfied: six>=1.5 in c:\users\hp\anaconda3\lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.16.0)

Requirement already satisfied: MarkupSafe>=2.1.1 in c:\users\hp\anaconda3\lib\site-packages (from Werkzeug<3.1->dash) (2.1.1)

Requirement already satisfied: zipp>=0.5 in c:\users\hp\anaconda3\lib\site-packages (from importlib-metadata->dash) (3.11.0)

Requirement already satisfied: comm>=0.1.1 in c:\users\hp\anaconda3\lib\site-packages (from ipykernel->jupyter-dash) (0.1.2)

Requirement already satisfied: debugpy>=1.6.5 in c:\users\hp\anaconda3\lib\site-packages (from ipykernel->jupyter-dash) (1.6.7)

Requirement already satisfied: jupyter-client>=6.1.12 in c:\users\hp\anaconda3\lib\site-packages (from ipykernel->jupyter-dash) (7.4.9)

Requirement already satisfied: jupyter-core!=5.0.*,>=4.12 in c:\users\hp\anaconda3\lib\site-packages (from ipykernel->jupyter-dash) (5.3.0)

Requirement already satisfied: matplotlib-inline>=0.1 in c:\users\hp\anaconda3\lib\site-packages (from ipykernel->jupyter-dash) (0.1.6)

Requirement already satisfied: packaging in c:\users\hp\anaconda3\lib\site-packages (from ipykernel->jupyter-dash) (23.1)

Requirement already satisfied: psutil in c:\users\hp\anaconda3\lib\site-packages (from ipykernel->jupyter-dash) (5.9.0)

Requirement already satisfied: pyzmq>=20 in c:\users\hp\anaconda3\lib\site-packages (from ipykernel->jupyter-dash) (23.2.0)

Requirement already satisfied: tornado>=6.1 in c:\users\hp\anaconda3\lib\site-packages (from ipykernel->jupyter-dash) (6.3.2)

Requirement already satisfied: traitlets>=5.4.0 in c:\users\hp\anaconda3\lib\site-packages (from ipykernel->jupyter-dash) (5.7.1)

Requirement already satisfied: backcall in c:\users\hp\anaconda3\lib\site-packages (from ipython->jupyter-dash) (0.2.0)

Requirement already satisfied: decorator in c:\users\hp\anaconda3\lib\site-packages (from ipython->jupyter-dash) (5.1.1)

Requirement already satisfied: jedi>=0.16 in c:\users\hp\anaconda3\lib\site-packages (from ipython->jupyter-dash) (0.18.1)

Requirement already satisfied: pickleshare in c:\users\hp\anaconda3\lib\site-packages (from ipython->jupyter-dash) (0.7.5)

Requirement already satisfied: prompt-toolkit!=3.0.37,<3.1.0,>=3.0.30 in c:\users\hp\anaconda3\lib\site-packages (from ipython->jupyter-dash) (3.0.36)

Requirement already satisfied: pygments>=2.4.0 in c:\users\hp\anaconda3\lib\site-packages (from ipython->jupyter-dash) (2.15.1)

Requirement already satisfied: stack-data in c:\users\hp\anaconda3\lib\site-packages (from ipython->jupyter-dash) (0.2.0)

Requirement already satisfied: colorama in c:\users\hp\anaconda3\lib\site-packages (from ipython->jupyter-dash) (0.4.6)

Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\hp\anaconda3\lib\site-packages (from requests->jupyter-dash) (2.0.4)

Requirement already satisfied: idna<4,>=2.5 in c:\users\hp\anaconda3\lib\site-packages (from requests->jupyter-dash) (3.4)

Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\hp\anaconda3\lib\site-packages (from requests->jupyter-dash) (1.26.16)

Requirement already satisfied: certifi>=2017.4.17 in c:\users\hp\anaconda3\lib\site-packages (from requests->jupyter-dash) (2023.7.22)

Requirement already satisfied: parso<0.9.0,>=0.8.0 in c:\users\hp\anaconda3\lib\site-packages (from jedi>=0.16->ipython->jupyter-dash) (0.8.3)

Requirement already satisfied: entrypoints in c:\users\hp\anaconda3\lib\site-packages (from jupyter-client>=6.1.12->ipykernel->jupyter-dash) (0.4)
Requirement already satisfied: platformdirs>=2.5 in c:\users\hp\anaconda3\lib\site-packages (from jupyter-core!=5.0.*,>=4.12->ipykernel->jupyter-dash) (3.10.0)
Requirement already satisfied: pywin32>=300 in c:\users\hp\anaconda3\lib\site-packages (from jupyter-core!=5.0.*,>=4.12->ipykernel->jupyter-dash) (305.1)
Requirement already satisfied: wcwidth in c:\users\hp\anaconda3\lib\site-packages (from prompt-toolkit!=3.0.37,<3.1.0,>=3.0.30->ipython->jupyter-dash) (0.2.5)
Requirement already satisfied: executing in c:\users\hp\anaconda3\lib\site-packages (from stack-data->ipython->jupyter-dash) (0.8.3)
Requirement already satisfied: asttokens in c:\users\hp\anaconda3\lib\site-packages (from stack-data->ipython->jupyter-dash) (2.0.5)
Requirement already satisfied: pure-eval in c:\users\hp\anaconda3\lib\site-packages (from stack-data->ipython->jupyter-dash) (0.2.2)
Note: you may need to restart the kernel to use updated packages.

```
In [1]: # 📌 1. Imports
import pandas as pd
from dash import Dash
from dash import dcc, html, Input, Output
import plotly.express as px
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder

# 📌 2. Load and clean your data
df = pd.read_excel("destiny edutech data.xlsx")
df.columns = df.columns.str.strip()
df.dropna(subset=['Course Name', 'Final Score (%)', 'Drop-off Status'], inplace=True)

# 📌 3. Encode for model
le_course = LabelEncoder()
le_status = LabelEncoder()

df['Course Code'] = le_course.fit_transform(df['Course Name'])
df['Drop-off Encoded'] = le_status.fit_transform(df['Drop-off Status'])

# 📌 4. Train the model
features = ['Course Code', 'Final Score (%)', 'Number of Attempts', 'Student F
X = df[features]
y = df['Drop-off Encoded']

model = RandomForestClassifier()
model.fit(X, y)
```

```
Out[1]: ▾ RandomForestClassifier
RandomForestClassifier()
```

```
In [2]: app = Dash(__name__)
app.layout = html.Div([
    html.H2(" Destiny Edutech Engagement Dashboard"),

    html.H4("Drop-off Predictor"),
    html.Label("Select Course"),
    dcc.Dropdown(id='course', options=[{'label': i, 'value': i} for i in df['C
        value=df['Course Name'].unique()[0]),

    html.Label("Final Score (%)"),
    dcc.Slider(0, 100, step=1, value=60, id='score'),

    html.Label("Number of Attempts"),
    dcc.Slider(1, 10, step=1, value=2, id='attempts'),

    html.Label("Student Feedback (1-10)"),
    dcc.Slider(1, 10, step=1, value=5, id='feedback'),

    html.Label("Days to Complete"),
    dcc.Slider(1, 100, step=1, value=30, id='days'),

    html.Br(),
    html.Div(id='prediction'),

    html.Hr(),
    html.H4(" Score Analyzer"),
    dcc.Graph(id='score-distribution'),

    html.Hr(),
    html.H4(" Dynamic Visualizations"),
    dcc.Graph(id='feedback-dist'),
    dcc.Graph(id='dropoff-chart'),
    dcc.Graph(id='days-box')
])
```

In [3]: # 📌 6. Callbacks for interactivity

```
@app.callback(
    Output('prediction', 'children'),
    Output('score-distribution', 'figure'),
    Output('feedback-dist', 'figure'),
    Output('dropoff-chart', 'figure'),
    Output('days-box', 'figure'),
    Input('course', 'value'),
    Input('score', 'value'),
    Input('attempts', 'value'),
    Input('feedback', 'value'),
    Input('days', 'value'),
)
def update_dashboard(course, score, attempts, feedback, days):
    course_code = le_course.transform([course])[0]
    pred = model.predict([[course_code, score, attempts, feedback, days]])
    pred_label = le_status.inverse_transform(pred)[0]

    filtered = df[df['Course Name'] == course]

    fig1 = px.histogram(filtered, x='Final Score (%)', nbins=20, title=f"Final Score Distribution")
    fig2 = px.histogram(filtered, x='Student Feedback (1-10)', nbins=10, title=f"Student Feedback Distribution")

    status_count = filtered['Drop-off Status'].value_counts().reset_index()
    status_count.columns = ['Status', 'Count']
    fig3 = px.bar(status_count, x='Status', y='Count', title="Drop-off vs Completed")

    fig4 = px.box(filtered, y='Days to Complete', title="Days to Complete (Box Plot)")

    return f"🎯 Predicted Outcome: {pred_label}", fig1, fig2, fig3, fig4
```

```
In [4]: app.run_server(mode='external', port=8054)
```

Destiny Edutech Engagement Dashboard

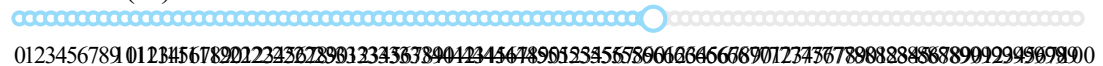
Drop-off Predictor

Select Course

Web Dev Bootcamp



Final Score (%)



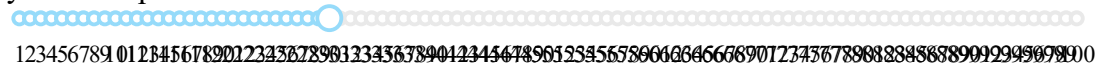
Number of Attempts



Student Feedback (1–10)



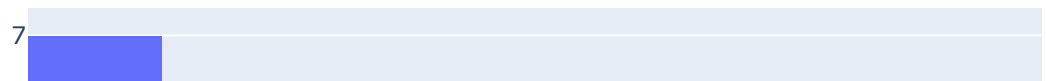
Days to Complete



 Predicted Outcome: No

Score Analyzer

Final Score Distribution: Web Dev Bootcamp



C:\Users\HP\anaconda3\Lib\site-packages\sklearn\base.py:464: UserWarning:

```
X does not have valid feature names, but RandomForestClassifier was fitted with feature names
```

In []:

